

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.		
Student Name:	S Dharshini	Reg. No.:	192511260
Section / Batch:	CSE -2025	Date:	03/08/2026

Code of Conduct

I, S Dharshini
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: S Dharshini

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:			100 Marks

Traversal Challenge

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Input

Input is level-by-level order.

10,20,30,100,150,200,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

10 20 100 150 30 200 300

10

20

30

100

150

200

300

Back to Home

Tree Traversals

Input

Input is level-by-level order.

10,20,30,100,150,200,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

100 20 150 10 200 30 300

10

20

30

100

150

200

300

Back to Home

Tree Traversals

Input

Input is level-by-level order.

10,20,30,100,150,200,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

100 150 20 200 300 30 10

10

20

30

100

150

200

300

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Binary Search Tree

Input

Input an array to convert to a BST.

7, 5, 1, 8, 3, 6, 0, 9, 4, 2

Enter numbers separated by commas. e.g., 8, 3, 10, 1, 6, 14

Visualize

Balance the Tree

Generate Python Code

```
graph TD; 7((7)) --> 5((5)); 7 --> 8((8)); 5 --> 1((1)); 5 --> 6((6)); 1 --> 0((0)); 1 --> 3((3)); 3 --> 2((2)); 3 --> 4((4)); 8 --> 9((9))
```

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Binary Search Tree

Input

Input an array to convert to a BST.

5, 2, 8, 1, 4, 7, 9, 0, null, 3, null, 6

Enter numbers separated by commas. e.g., 8, 3, 10, 1, 6, 14

Visualize

Balance the Tree

Generate Python Code

```
graph TD; 5((5)) --> 2((2)); 5 --> 8((8)); 2 --> 1((1)); 2 --> 4((4)); 1 --> 0((0)); 4 --> 3((3)); 8 --> 7((7)); 8 --> 9((9)); 7 --> 6((6))
```

4. C) A, B, C, D, E, F, G, H, I

5. 1. Pre-order Traversal (Root → Children)

- Visit Root: **A**
- Traverse subtree rooted at **B**: **B, E, F, I, J, K**
- Traverse subtree rooted at **C**: **C, G, H**
- Traverse subtree rooted at **D**: **D**
- **Sequence: A, B, E, F, I, J, K, C, G, H, D**

2. Post-order Traversal (Children → Root)

- Left subtree children first: **E, I, J, K, F, B**
- Middle subtree children: **G, H, C**
- Right subtree node: **D**
- Root node last: **A**
- **Sequence: E, I, J, K, F, B, G, H, C, D, A**

3. In-order Traversal (for multi-way / general trees, typically Leftmost Child → Root → Remaining Children)

- **Sequence: E, B, I, F, J, K, A, G, C, H, D**

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

1245367

1

2

3

4

5

6

7

← Back to Home

Tree Traversals

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

4251637

1

2

3

4

5

6

7

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

4526731

1

2

3

4

5

6

7